Boundary Value Analysis Report

Document control

Field	Value
Student/group	10
Date updated	2026-10-19
Feature completed in this step	FR-05, FR-11, and FR-20 sections
Precondition	Gate A approved for FR-05, FR-11, and FR-20

BVA rules applied

- The authoritative requirement source is the project `README.md`.
- No maximum search length or maximum result count is specified in `README.md`, the inspected UI, or the inspected product endpoint.
- Therefore no arbitrary upper boundary is introduced. It is recorded as `Unresolved`.
- Each BVA case changes one variable/domain only. Other data remains nominal.
- For a categorical set such as matching/no-match, BVA is not forced where there is no ordered numeric boundary; the observed result-count transition is used only where it has a meaningful `0/1/multiple` boundary.

Evidence used

Evidence ID	Source	Observation
BVA-E-01	Project `README.md`, FR-05	Home page displays all products, searches by product name, shows loading, and shows an empty state for no results.
BVA-E-02	`frontend-web/src/pages/Home.jsx`	Empty search is sent to the product endpoint; non-empty search is submitted; product result count is rendered when results are non-empty.
BVA-E-03	`backend/server.js`, `GET /api/products?search=...`	Empty/falsy `search` returns all products; non-empty search returns rows matching the product-name `LIKE` query.
BVA-E-04	Current seeded database	Five products are available; one verified name can produce one result, and an absent term can produce zero results.

1. Ordered domains and boundaries

1.1 Search input length: empty versus non-empty

The requirement distinguishes an empty search from a non-empty search, but does not define a maximum length.

BVA ID	Variable/domain	LB	UB	Alpha	Nominal	Smallest/largest UI value	Boundary points	Expected observable output	Evidence	Review
BVA-FR05-01	Search text length	0 characters	Unresolved	1 character	""	Smallest: 0; largest: Unresolved	0, 1	At 0, all products are requested/displayed. At 1, a non-empty search request is sent; output depends on matches.	BVA-E-01,BVA-E-02,BVA-E-03	Approv

`LB-alpha` would be `-1` character and is not an executable UI input. It is therefore excluded and documented rather than fabricated. `UB`, `UB-alpha`, and `UB+alpha` cannot be generated until a real maximum is specified or verified.

1.2 Product result cardinality

The feature requirement and current seed data support the observable result classes `0`, `1`, and `multiple`. This is a result cardinality domain, not a user-entered numeric field.

BVA ID	Variable/domain	LB	UB	Alpha	Nominal	Boundary points	Expected observable output	Evidence	Review
BVA-FR05-02	Search result count	0 results	Unresolved for multiple result upper limit	1 result	1 verified match	0, 1, 2+ observed results	0: empty state; 1: one matching product; 2+: multiple matching products/list cards.	BVA-E-01,BVA-E-02,BVA-E-04	Approved

There is no specified maximum number of products or pagination boundary in the available evidence. `2+` is therefore an observed multi-result class, not an invented upper boundary.

2. Boundary Value Test Cases

All cases vary one FR-05 variable only. Product data, request availability, and the UI surface remain nominal unless the row explicitly varies the request state.

Test ID	Variable varied	Input/controlled data	Other variables at nominal values	Expected observable result	Actual result	Status	Traceability
BVA-TC-FR05-01	Search text length	Empty string `""` (0 characters)	Product API available; seeded product data	All available products are requested and product cards/list are displayed.	Screenshot shows five product cards after empty search.	Passed	FR05-EC-01, BVA-FR05-01; FR05-02-searchEmpty.png (../../screenshots/FR-05/FR05-02-searchEmpty.png)
BVA-TC-FR05-02	Search text length	One-character non-empty term, e.g. `i`	Product API available; same seeded product data	A non-empty search request is sent; the returned result cardinality is recorded without assuming a specific count.	Screenshot shows term `i` and two matching product cards.	Passed	FR05-EC-02/03, BVA-FR05-01; FR05-03-searchOneChar.png (../../screenshots/FR-05/FR05-03-searchOneChar.png)
BVA-TC-FR05-03	Result cardinality	Search term producing 0 results	Same UI/API and nominal request state	Empty-result state is observable and no matching product card is displayed.	Screenshot shows no product card for `xyz-no-product-999`.	Passed	FR05-EC-03, BVA-FR05-02; FR05-04-searchNoResult.png (../../screenshots/FR-05/FR05-04-searchNoResult.png)
BVA-TC-FR05-04	Result cardinality	Search term producing exactly 1 result	Same UI/API and nominal request state	Exactly one matching product is displayed.	Screenshot shows one product card for `iPhone 15`.	Passed	FR05-EC-02, BVA-FR05-02; FR05-05-searchOneResult.png (../../screenshots/FR-05/FR05-05-searchOneResult.png)
BVA-TC-FR05-05	Result cardinality	Search term producing 2 or more results	Same UI/API and nominal request state	Multiple matching products are displayed.	Screenshot shows two product cards for `1`.	Passed	FR05-EC-02, BVA-FR05-02; FR05-06-searchTwo+Result.png (../../screenshots/FR-05/FR05-06-searchTwo+Result.png)

3. Boundary limitations

Potential boundary	Status	Reason
Maximum search string length	Unresolved	Not specified in project `README.md`, UI source, or endpoint contract inspected.
Allowed special-character length/format	Unresolved	No explicit input constraint was found; do not invent one.
Maximum product result count/pagination	Unresolved	No pagination or maximum result count was specified or observed.
Loading timing threshold	Unresolved	No measurable response-time threshold is specified.

Gate B review

Review item	Status	Student correction/confirmation
Every justified ordered domain has LB/UB/alpha	Approved	Student approved on 2026-10-19; no corrections
Unspecified limits are not invented	Approved	Student approved on 2026-10-19; no corrections
One-variable-at-a-time constraint satisfied	Approved	Student approved on 2026-10-19; no corrections
Gate B approved before Step C	Approved	Student approved on 2026-10-19; UI evidence for Step C will be provided later

4. FR-11 - User order history

Evidence used

Evidence ID	Source	Observation
BVA-FR11-E-01	Project `README.md`, FR-11	Users can view only their own order history.
BVA-FR11-E-02	`frontend-web/src/pages/Profile.jsx`	History renders order ID, date, total, translated status, and an empty-order message.
BVA-FR11-E-03	`backend/server.js`, `GET /api/orders/my-orders`	Authenticated requests are filtered by `req.user.id` and sorted by descending order ID.
BVA-FR11-E-04	`backend/database.js`, orders schema	Orders use integer IDs and a `pending` default status; no maximum count is specified.

BVA rules and boundary analysis

The evidence-backed ordered domain is order-history cardinality:

- LB: `0` orders.
- Alpha: `1` order.
- Observed multiple-order class: `2+` orders.
- UB: unresolved; no maximum order count or pagination limit is specified.
- `LB-alpha` is `-1` order and is not executable.

Authentication (`valid`, `missing`, `invalid`) and order status (`pending`, `confirmed`, `shipping`, `delivered`, `canceled`) are categorical/state domains. They remain equivalence-partition cases and are not assigned artificial numeric boundaries.

BVA ID	Variable/domain	LB	UB	Alpha	Nominal	Boundary points	Expected observable output	Evidence	Review
BVA-FR11-01	Order count	0 orders	Unresolved	1 order	1 own order	0, 1, 2+ observed orders	At 0, the empty-order message is displayed. At 1, one order is displayed. At 2+, multiple own orders are displayed in descending ID order.	BVA-FR11-E-01 through BVA-FR11-E-04	Pending

Boundary Value Test Cases

Test ID	Variable varied	Input/controlled data	Other variables at nominal values	Expected observable result	Actual result	Status	Traceability
BVA-TC-FR11-01	Order count	User A with 0 orders	Authenticated User A; order-history endpoint available	Empty-order message is displayed.	UI evidence confirms the empty-order state.	Passed	FR11-EC-01/03, BVA-FR11-01; FR-11-03-empty.png (../../screenshots/FR-11/FR-11-03-empty.png)
BVA-TC-FR11-02	Order count	User A with exactly 1 own order	Authenticated User A; one own order	One order with ID, date, total, and translated status is displayed.	UI evidence confirms the one-order history state.	Passed	FR11-EC-01/04, BVA-FR11-01; FR-11-01-exactOneOrder.png (../../screenshots/FR-11/FR-11-01-exactOneOrder.png)
BVA-TC-FR11-03	Order count	User A with 2+ own orders	Authenticated User A; multiple own orders	All own orders are displayed in descending ID order.	UI evidence confirms the multiple-order history state.	Passed	FR11-EC-01/05, BVA-FR11-01; FR-11-02-multiOrder.png (../../screenshots/FR-11/FR-11-02-multiOrder.png)

Additional FR-11 evidence filenames supplied by the student:

- FR-11-02-multiOrderSupported.png (../../screenshots/FR-11/FR-11-02-multiOrderSupported.png)
  - supplementary multi-order/status evidence;
- FR-11-04-missingOrInvalidToken.png (../../screenshots/FR-11/FR-11-04-missingOrInvalidToken.png)
  - missing/invalid-token evidence;
- FR11-05-userAOrders.png (../../screenshots/FR-11/FR11-05-userAOrders.png) and FR11-05-userBOrders.png (../../screenshots/FR-11/FR11-05-userBOrders.png)
  - User A/User B ownership evidence.

Boundary limitations

Potential boundary	Status	Reason
Maximum order count/pagination	Unresolved	No upper limit is specified or observed.
Order ID minimum/maximum	Unresolved	Integer primary key exists, but no user-facing ID range is specified.
Authentication token boundary	Not numeric	Valid, missing, and invalid are state classes.
Order status ordering	Not numeric	Supported status values are categorical.

Gate B review

Review item	Status	Student correction/confirmation
Every justified ordered domain has LB/UB/alpha	Approved	`Student approved on 2026-10-19; no corrections; 0/1/2+ order-count boundary documented`
Unspecified limits are not invented	Approved	`Student approved on 2026-10-19; maximum order count/pagination and order-ID range remain unresolved`
One-variable-at-a-time constraint satisfied	Approved	`Student approved on 2026-10-19; each BVA case varies order count only`
Gate B approved before Step C	Approved	`Student approved on 2026-10-19; no corrections`

5. FR-19 - Admin user management

Evidence used

Evidence ID	Source	Observation
BVA-FR19-E-01	Project `README.md`, FR-19	Admin can view all users without exposing passwords and may delete users except the currently logged-in administrator.

Evidence ID	Source	Observation
BVA-FR19-E-02	`frontend-admin/src/App.jsx`	Admin UI loads users and sends delete requests using the selected user ID.
BVA-FR19-E-03	`backend/server.js`, `/api/admin/users`	User listing selects user fields without password; delete endpoint accepts `:id` and uses authentication middleware.
BVA-FR19-E-04	`backend/database.js`, users schema	User IDs are integer primary keys; seeded dataset contains an admin and a normal user.

BVA rules and boundary analysis

Customer-user-list cardinality is an observable count domain with classes `0`, `1`, and `2+` customer users, while the logged-in admin remains a baseline record. No maximum list size or pagination limit is specified.

Target user identity has an observed integer-ID representation, but no user-facing minimum/maximum or malformed-ID contract is specified. Therefore no arbitrary numeric upper boundary is introduced.

Authentication/role and current-admin protection are categorical/security states, not numeric boundaries. The source does not establish that role or current-admin checks are enforced by the backend, so those outcomes remain verification targets.

BVA ID	Variable/domain	LB	UB	Alpha	Nominal	Boundary points	Expected observable output	Evidence	Review
BVA-FR19-01	Customer-user-list count	0 customer users	Unresolved	1 customer user	1 listed customer user	0, 1, 2+ customer users	At 0, the admin baseline is listed with no customer user. At 1, one customer user is listed. At 2+, multiple customer users are listed and password values are absent.	BVA-FR19-E-01 through BVA-FR19-E-04	Pending
BVA-FR19-02	Target user ID	Unresolved	Unresolved	1 integer ID	Existing other-user ID	Existing ID; current-admin ID; absent ID; malformed ID	Existing other-user ID triggers the delete flow; current-admin, absent, and malformed IDs require observed protection/error behavior.	BVA-FR19-E-02 through BVA-FR19-E-04	Pending

Boundary Value Test Cases

Test ID	Variable varied	Input/controlled data	Other variables at nominal values	Expected observable result	Actual result	Status	Traceability
BVA-TC-FR19-01	Customer-user-list count	Admin with 0 customer users	Admin authentication; user-management route available	Admin baseline remains listed and no customer user is listed.	UI evidence filename supplied by student.	Passed	FR19-EC-01/04, BVA-FR19-01; FR-19-01-zeroUser.png (../screenshots/FR-19/FR-19-01-zeroUser.png)
BVA-TC-FR19-02	Customer-user-list count	Admin with exactly 1 customer user	Admin authentication; one valid customer user record	Admin and one customer user are listed; password is not exposed.	UI evidence filename supplied by student.	Passed	FR19-EC-01/05, BVA-FR19-01; FR-19-02-oneUser.png (../screenshots/FR-19/FR-19-02-oneUser.png)
BVA-TC-FR19-03	Customer-user-list count	Admin with 2+ customer users	Admin authentication; multiple valid customer user records	Admin and multiple customer users are listed; password values are absent.	UI evidence filename supplied by student.	Passed	FR19-EC-01/05, BVA-FR19-01; FR-19-03-multipleUser.png (../screenshots/FR-19/FR-19-03-multipleUser.png)
BVA-TC-FR19-04	Target user ID	Existing other-user integer ID `3`	Admin authentication; target is not current admin	The selected other user is deleted and the refreshed list no longer contains that user.	UI evidence filename supplied by student.	Passed	FR19-EC-01/06, BVA-FR19-02; FR-19-04-delUserID=3.png (../screenshots/FR-19/FR-19-04-delUserID=3.png)
BVA-TC-FR19-05	Target user ID	Current-admin integer ID	Admin authentication; no other target variation	Current administrator remains in the user list and current admin access is protected.	UI shows deletion success; after refresh admin is absent from the list while the existing session remains active.	Failed	FR19-EC-01/07, BVA-FR19-02; FR-19-05-delAdminUserSuccessfully.png (../screenshots/FR-19/FR-19-05-delAdminUserSuccessfully.png); `BUG-FR19-001`

Test ID	Variable varied	Input/controlled data	Other variables at nominal values	Expected observable result	Actual result	Status	Traceability
BVA-TC-FR19-06	Target user ID	Absent integer ID and malformed IDs	Admin authentication; one invalid identifier per request	Invalid identifiers produce an error or defined no-op response without false success.	Requests return `200 OK` with `{"message":"User deleted"}` for absent/malformed IDs; valid user list remains unchanged.	Failed	FR19-EC-08, BVA-FR19-02; FR-19-06-deleteInvalidUser.md (../screenshots/FR-19/FR-19-06-deleteInvalidUser.md); `BUG-FR19-002`

Boundary limitations

Potential boundary	Status	Reason
Maximum user-list count/pagination	Unresolved	No upper limit is specified or observed.
User ID minimum/maximum	Unresolved	Integer primary key exists, but no user-facing range is specified.
Malformed identifier format	Unresolved	Endpoint parameter format and expected error are not defined by the checked evidence.
Role/current-admin authorization boundary	State/security domain	Must be verified at runtime; source does not prove the required checks.

Gate B review

Review item	Status	Student correction/confirmation
Every justified ordered domain has LB/UB/alpha	Approved	`Student approved on 2026-10-20; no corrections; 0/1/2+ user-list boundary documented`
Unspecified limits are not invented	Approved	`Student approved on 2026-10-20; list size/pagination and user-ID range remain unresolved`
One-variable-at-a-time constraint satisfied	Approved	`Student approved on 2026-10-20; each case varies one domain only`
Gate B approved before Step C	Approved	`Student approved on 2026-10-20; no corrections`

6. FR-20 - Mobile product browsing and order history

Scope and evidence used

FR-20 is limited to mobile product browsing/search and authenticated mobile order history. Mobile authentication, profile editing, cart, checkout, and order cancellation are out of scope for this BVA section.

Evidence ID	Source	Observation
BVA-FR20-E-01	Project `README.md`, FR-20 and product-listing requirements	Mobile includes product viewing and order history; product search distinguishes loading and empty-result states.
BVA-FR20-E-02	`frontend-mobile/App.js`, product loading/search logic	Empty search is submitted; non-empty search is URL-encoded; loading, product-list, empty-list, and error states are represented.
BVA-FR20-E-03	`frontend-mobile/App.js`, order-history logic	Authenticated order history is requested for the current token and parsed as an order list.
BVA-FR20-E-04	FR-05 and FR-11 approved BVA sections	Product result cardinality uses `0`, `1`, and `2+`; order-history cardinality uses `0`, `1`, and `2+`.

BVA rules and boundary analysis

The ordered mobile search boundary is the same observed empty/non-empty boundary as FR-05. The product result and order-history cardinalities reuse the approved `0/1/2+` classes from FR-05 and FR-11. Loading/error, token authentication, and ownership are categorical or security states rather than numeric boundaries.

No maximum search length, maximum product-result count, maximum order count, pagination limit, or order-ID range is specified. These limits remain `Unresolved`; no artificial upper boundary is introduced.

BVA ID	Variable/domain	LB	UB	Alpha	Nominal	Boundary points	Expected observable output	Evidence	Review
BVA-FR20-01	Mobile search text length	0 characters	Unresolved	1 character	Empty search ""	0, 1	At 0, the mobile product list is requested/displayed. At 1, a non-empty search is sent and its result cardinality is observed.	BVA-FR20-E-01 through BVA-FR20-E-02	Pending

BVA ID	Variable/domain	LB	UB	Alpha	Nominal	Boundary points	Expected observable output	Evidence	Review
BVA-FR20-02	Mobile product result count	0 results	Unresolved for maximum result count	One result	matching product	0, 1, 2+	0 shows an empty state; 1 shows one matching product; 2+ shows multiple product cards.	BVA-FR20-E-01, BVA-FR20-E-02, BVA-FR20-E-04	Pending
BVA-FR20-03	Mobile order-history count	0 own orders	Unresolved for maximum order count	1 own order	One own order	0, 1, 2+	0 shows the empty history state; 1 shows one own order; 2+ shows all own orders with required fields/status labels.	BVA-FR20-E-01, BVA-FR20-E-03, BVA-FR20-E-04	Pending

Boundary Value Test Cases

All cases vary one numeric/cardinality domain only. Authentication, API availability, and other data remain nominal unless the row explicitly identifies a state/security check.

Test ID	Variable varied	Input/controlled data	Other variables at nominal values	Expected observable result	Actual result	Status	Traceability
BVA-TC-FR20-01	Mobile search text length	Empty search `""` (0 characters)	Product API available; nominal mobile product data	Mobile requests/displays the product list.	API returned HTTP 200 with 5 products; mobile UI still needs visual evidence.	API Passed; UI pending	FR20-EC-01, BVA-FR20-01; student script output 2026-10-20
BVA-TC-FR20-02	Mobile search text length	One-character non-empty search term `i`	Product API available; same nominal product data	Mobile sends a non-empty search request and displays the returned result state.	API returned HTTP 200 with 2 products; mobile UI still needs visual evidence.	API Passed; UI pending	FR20-EC-02/03, BVA-FR20-01; student script output 2026-10-20
BVA-TC-FR20-03	Mobile product result count	Search term `fr20-no-product-999999` producing 0 results	Nominal mobile request state	Empty product-result state is displayed and no matching product card is shown.	API returned HTTP 200 with 0 products; mobile empty-state UI still needs visual evidence.	API Passed; UI pending	FR20-EC-03, BVA-FR20-02; student script output 2026-10-20
BVA-TC-FR20-04	Mobile product result count	Search term `iPhone 15 Pro Max` producing exactly 1 result	Nominal mobile request state	Exactly one matching product card is displayed.	API returned HTTP 200 with 1 product; mobile UI still needs visual evidence.	API Passed; UI pending	FR20-EC-02, BVA-FR20-02; student script output 2026-10-20
BVA-TC-FR20-05	Mobile product result count	Search term `a` producing 2+ results	Nominal mobile request state	Multiple matching product cards are displayed.	API returned HTTP 200 with 4 products; mobile UI still needs visual evidence.	API Passed; UI pending	FR20-EC-02, BVA-FR20-02; student script output 2026-10-20
BVA-TC-FR20-09	Mobile product request state	Controlled delayed product request	Nominal mobile product screen	`Đang tải...` is displayed while the product request is pending.	Mobile UI displayed `Đang tải...` in the product-list block.	Passed	FR20-EC-04, BVA-FR20-02; TC-FR20-MOB-006.png (../screenshots/FR-20/TC-FR20-MOB-006.png)
BVA-TC-FR20-10	Mobile product request state	Controlled failed product request	Nominal mobile product screen	A visible error state is displayed after the request fails.	Mobile UI displayed `Failed to fetch` at the upper-left of the product-list block.	Passed	FR20-EC-04, BVA-FR20-02; TC-FR20-MOB-007.png (../screenshots/FR-20/TC-FR20-MOB-007.png)
BVA-TC-FR20-06	Mobile order-history count	Authenticated user with 0 own orders	Valid token; order-history API available	Mobile displays the empty order-history state.	API returned HTTP 200 with `userId=2`, `count=0`; mobile UI still needs visual evidence.	API Passed; UI pending	FR20-EC-05/06, BVA-FR20-03; student script output 2026-10-20
BVA-TC-FR20-07	Mobile order-history count	Authenticated user with exactly 1 own order	Valid token; order-history API available	Mobile displays one own order with its required fields and status label.	With User ID 2 configured with exactly 1 order, API returned `PASS`, `userId=2`, `count=1`; mobile UI still needs visual evidence.	API Passed; UI pending	FR20-EC-05/06, BVA-FR20-03; student script output 2026-10-20
BVA-TC-FR20-08	Mobile order-history count	Authenticated user with 2+ own orders	Valid token; order-history API available	Mobile displays all own orders with their required fields and status labels.	With User ID 2 configured with 2 orders, API returned `PASS`, `userId=2`, `count=2`; mobile UI still needs visual evidence.	API Passed; UI pending	FR20-EC-05/06, BVA-FR20-03; student script output 2026-10-20

Boundary limitations and non-numeric states

Potential boundary/state	Status	Reason
Maximum mobile search length	Unresolved	No maximum is specified or observed.
Maximum product-result count/pagination	Unresolved	No upper result or pagination limit is specified.
Maximum order count/pagination	Unresolved	No upper order-history limit is specified.

Potential boundary/state	Status	Reason
Loading/error request state	State domain	Must be tested as a categorical request state, not a numeric boundary.
Authentication and order ownership	State/security domain	Must be tested with valid/invalid token and separate-user data; no numeric boundary is created.

Gate B review

Review item	Status	Student correction/confirmation
Every justified ordered domain has LB/UB/alpha	Approved	`Student approved on 2026-10-20; mobile search, product-result, and order-history boundaries reviewed with no corrections`
Unspecified limits are not invented	Approved	`Student approved on 2026-10-20; maximum lengths/counts and pagination remain unresolved`
One-variable-at-a-time constraint satisfied	Approved	`Student approved on 2026-10-20; each proposed BVA case varies one boundary domain only`
Gate B approved before Step C	Approved	`Student approved on 2026-10-20; no corrections`